

Basic Principles of Medicine 1
Module: Foundation to Medicine
Lecture No: CPR 35

Ventilation

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Note: Blue slides have the written explanation of the previous slides which will be discussed in the class. Therefore, the blue slides might not be included in the lecture time. However, all the learning materials (including the blue slides) should be read and understood for the complete grasp of the topic.

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Recommended Reading

Rhoades and Bell, *Medical Physiology: Principles for Clinical Medicine*, 6th Edition, © Copyright © 2023 Wolters Kluwer, Chapter 18

<https://meded-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=253088352&bookid=3211>

Learning Objectives

1. Define partial pressure and fractional concentration as they apply to gases in air. 2. List the normal fractional concentrations and sea level partial pressures for O ₂ , CO ₂ , and N ₂ .	SOM.MK.II.BPM1.3.CPR.2.P HYS.0232
List the normal airway, alveolar, arterial, and mixed venous PO ₂ and PCO ₂ , O ₂ saturation, [HCO ₃ ⁻] and pH.	SOM.MK.II.BPM1.3.CPR.2.P HYS.0233
Define and contrast the following terms: anatomic dead space, physiologic dead space, wasted (dead space) ventilation, total minute ventilation and alveolar minute ventilation.	SOM.MK.II.BPM1.3.CPR.2.P HYS.0234
Describe the concept by which physiological dead space can be measured.	SOM.MK.II.BPM1.3.CPR.2.P HYS.0235
Explain the relationship between alveolar ventilation and the arterial PCO ₂ and PO ₂ and pH, compliance and ventilation. Describe how hypoventilation would affect alveolar and arterial PO ₂ and PCO ₂ and the A-a gradient.	SOM.MK.II.BPM1.3.CPR.2.P HYS.0236

Learning Objectives

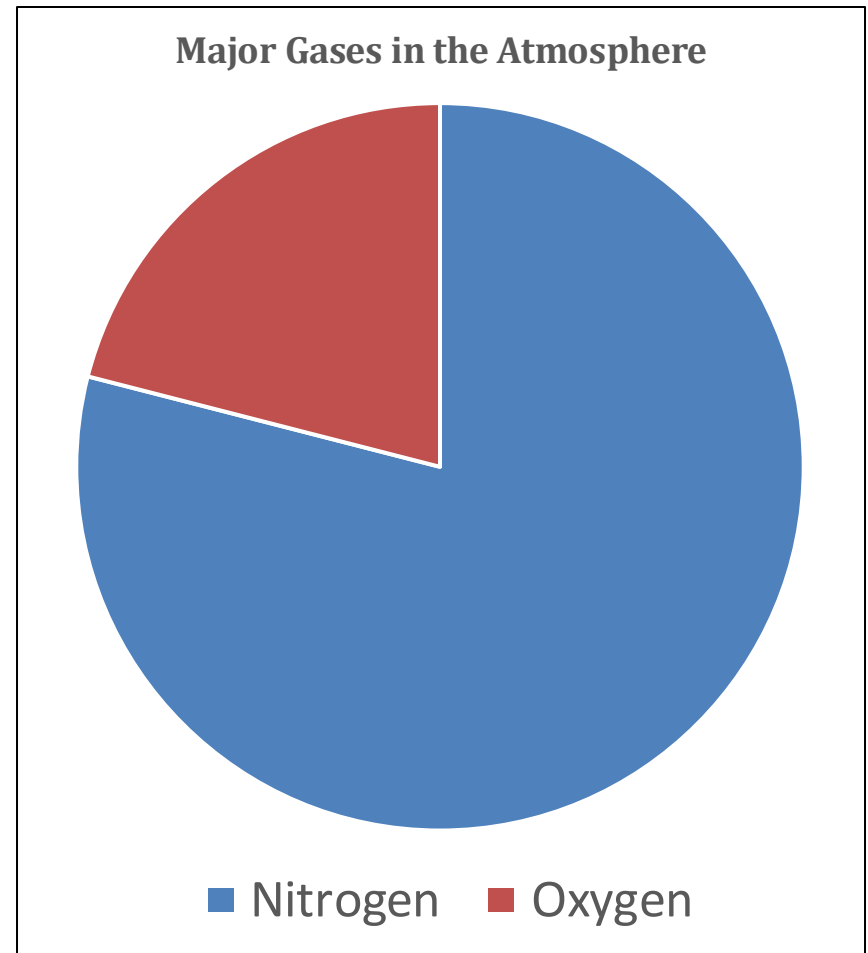
Describe in quantitative terms the effect of ventilation on PCO ₂ according to the alveolar ventilation equation.	SOM.MK.II.BPM1.3.CPR.2.PHYS .0237
Calculate the alveolar oxygen partial pressure (PAO ₂) and the amount of supplemental O ₂ required to overcome a reduction in PAO ₂ caused by hypoventilation or high altitude using the simplified form of the alveolar gas equation.	SOM.MK.II.BPM1.3.CPR.2.PHYS .0238
Define the following terms: hypoventilation, hyperventilation, hypercapnia, eupnea, hypopnea, and hyperpnea.	SOM.MK.II.BPM1.3.CPR.2.PHYS .0239
Calculate the alveolar to arterial PO ₂ difference (A-a) DO ₂ . List the normal value for (A-a) DO ₂ and explain the clinical significance of an elevated (A-a) DO ₂ .	SOM.MK.II.BPM1.3.CPR.2.PHYS .0240

Terms

- f – frequency
- F – fractional concentration
- E – expired
- I – inspired
- P_a – arterial partial pressure
- P_A – alveolar partial pressure
- P_{H_2O} – water vapor partial pressure
- P_B – barometric pressure
- V – volume
- V_T – tidal volume
- $\dot{V}_A$ – alveolar ventilation
- V_D – dead space volume

Partial Pressure

- **Partial pressure**
 - The pressure exerted by an individual gas in a mixture of known gases
 - Expressed in terms of its dry gas concentration
- **Fractional Concentration**
 - the percentage or concentration of a gas in a mixture of known gases



Partial Pressure in Ambient Air

Gas	Fractional Concentration %	Partial Pressure (mm Hg)
Nitrogen	79	600.4
Oxygen	21	159.6
Carbon Dioxide	0	0
Water vapor	0	0
Total	100	760

Composition of dry air in the atmosphere

Calculating partial pressure

Example

- $$P_{O_2} = P_B \times F_{O_2}$$
$$= 760 \times 21\%$$
$$= 159.6 \text{ mm Hg}$$

DIY case 1:

A 31-year-old marathon runner hikes to a remote mountain in South America twice a year as part of his training. This mountain is 2,810 m at its highest point and with a barometric pressure of 286 mm Hg. What is the PO_2 at this location?

- A. 50 mm Hg
- ✓ B. 60 mm Hg
- C. 159 mm Hg
- D. 286 mmHg
- E. 590 mmHg

Partial Pressure (In Trachea)

Gas	Fractional Concentration %	Partial Pressure (mmHg)
Nitrogen	74	563
Oxygen	20	150
Carbon Dioxide	0	0
Water	6	47
Total	100	760

Composition of air in the trachea

Calculating partial pressure

$$\begin{aligned}
 P_{IO_2} &= (P_B - P_{H_2O}) \times F_{IO_2} \\
 P_{IO_2} &= (760 - 47) \times 0.21 \\
 P_{IO_2} &= 149.7 \text{ mmHg}
 \end{aligned}$$

Air is humidified in the conducting zone therefore ; the partial pressure of water vapor (P_{H_2O}) must be taken into consideration.

Inspired P_{H_2O} is not altered by altitude.

Partial Pressure and Henry's Law

Table 18.2		Partial Pressures and Percentages of Respiratory Gases at sea level ($P_B = 760$ mmHg)						
Gas	Ambient Dry Air		Moist Tracheal Air		Alveolar Air		Arterial Blood	Venous Blood
	mm Hg	%	mm Hg	%	mm Hg	%	mm Hg	mm Hg
O ₂	160	21	150	20	102	14	95	40
CO ₂	0	0	0	0	40	5	40	46
Water vapor	0	0	47	6	47	6	47	47
N ₂	600	79	563	74	571	75	571	571
Total	760	100	760	100	760	100	760	704*

* – in venous blood the drop in PO₂ is greater than the rise in PCO₂

- Water vapor pressure depends on temperature, at 37 °C max 47 mm Hg.

DIY case 2

A 31-year-old marathon runner is brought to the team physician while on a hike to a remote mountain in South America because of a sudden onset of difficulty breathing (dyspnea). They are 2,810 m above sea level and the barometric pressure is 286 mm Hg. If he is given 40% O_2 via nasal cannula which of the following is the most likely PIO_2 of the patient?

- A. 50 mm Hg
- ✓ B. 95 mm Hg
- C. 114 mm Hg
- D. 159 mmHg
- E. 285 mmHg

Minute Ventilation

$$\dot{V}_E = VT \times f$$

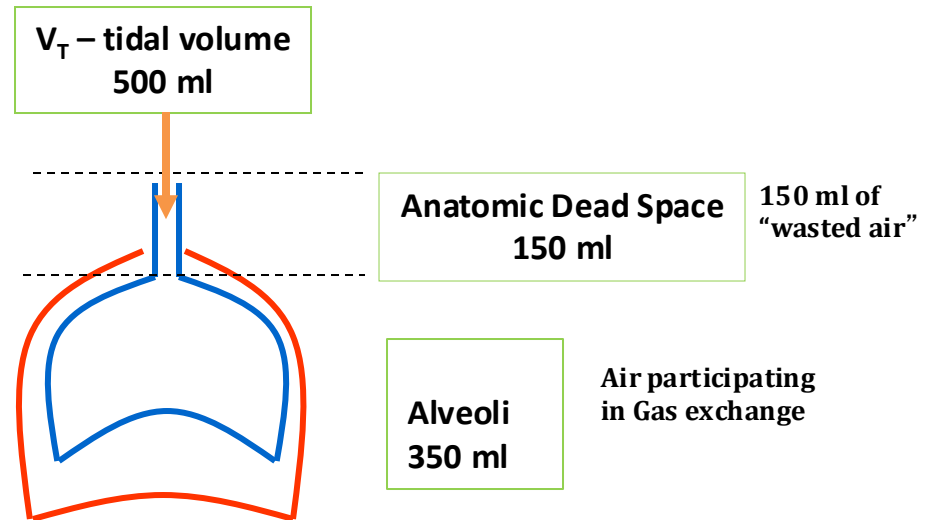
- Minute ventilation: volume of air entering the lungs per minute
- Expired minute ventilation ($\dot{V}_E$) : volume of expired air per minute, equal to minute ventilation
- V_T : tidal volume (typically ~ 500 ml)
- f : breathing frequency (typically ~ 12 breaths/min)
- With typical values:

$$\dot{V}_E = 0.5 \text{ liters} \times 12 \text{ min}^{-1} = 6 \text{ L/min}$$

Tidal Volume and Dead Space

Tidal volume is distributed into two compartments:

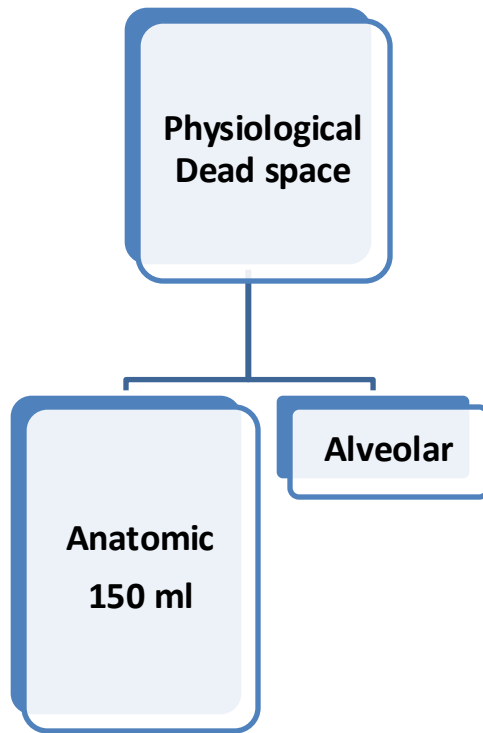
1. Dead space (V_D) in which gas exchange cannot occur;
2. Alveolar space where there is gas exchange



Wasted air: does not participate in gas exchange

The V_D / V_T ratio is usually between 0.25 and 0.35 in healthy individuals

Dead Space V_D



Physiological dead space =
anatomical V_D + alveolar V_D

In healthy individuals **Physiological V_D**
= anatomical V_D

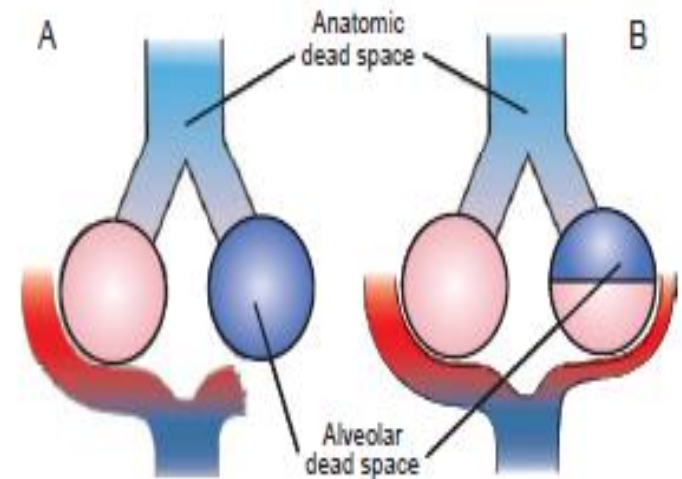
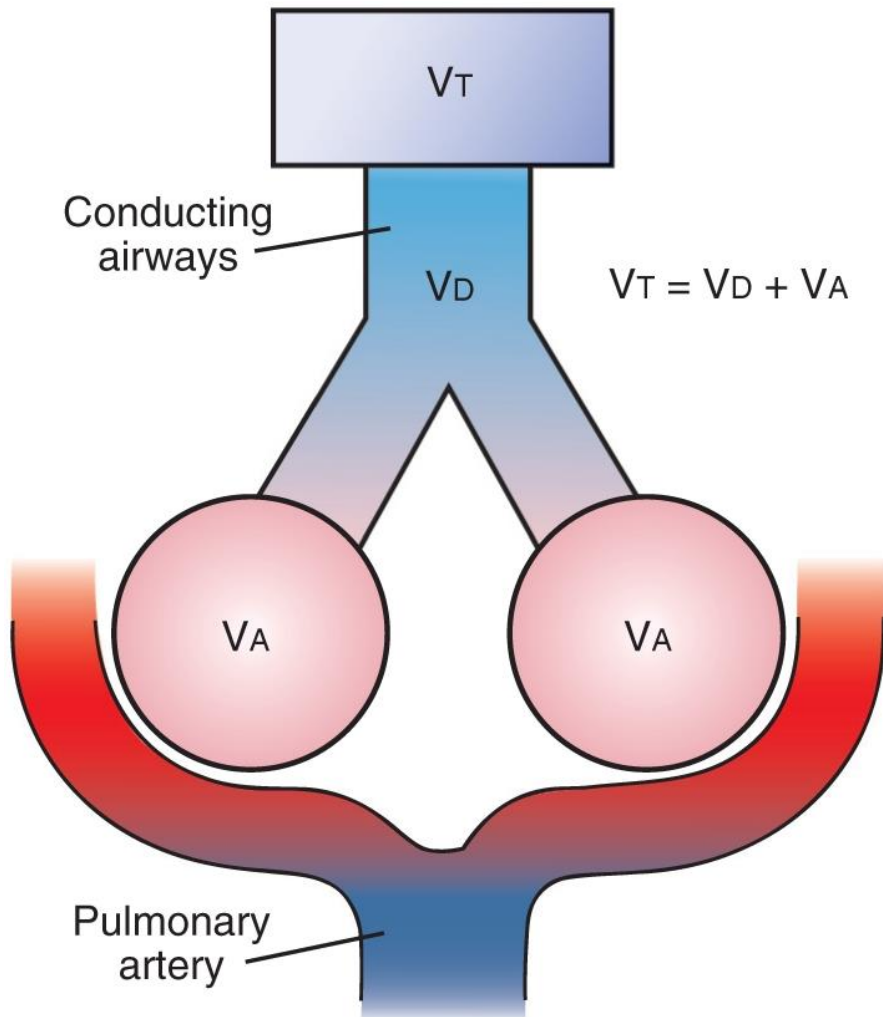


Figure 18.13 Total wasted air in the lungs is computed from the physiologic dead space volume. Dead space volume occurs in the conducting airways and in alveoli with poor capillary circulation. (A) There is no blood flow to an alveolar region. (B) There is reduced blood flow. In both cases, a portion of alveolar air does not participate in gas exchange and constitutes alveolar dead space volume. Note that physiologic dead space is the sum of alveolar dead space plus anatomic dead space.

Dead Space V_D

- Volume of inhaled air that does not participate in gas exchange
- Conducting airways – (**Anatomic dead space**)
- Alveoli with no perfusion - (**Alveolar dead space**), value is negligible in healthy individuals

Tidal Volume Components and Alveolar Ventilation



- V_T : is distributed in the $V_A + V_D$
- V_D – volume of air in the dead space
- V_A – volume of air added to the alveoli

Alveolar Ventilation

$$\dot{V}_A = (V_T \times f) - (V_D \times f)$$

Alveolar Ventilation ($\dot{V}_A$)

- Volume of air delivered to the respiratory zone per minute
- The fraction of inspired air that participates in gas exchange.

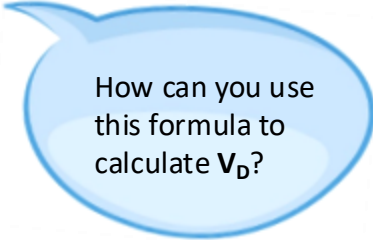
$$\dot{V}_A = (\dot{V}_T - V_D) \times \text{Breaths/min}$$

where

$\dot{V}_A$ = Alveolar ventilation (mL/min)

V_T = Tidal volume (mL)

V_D = Physiologic dead space (mL)



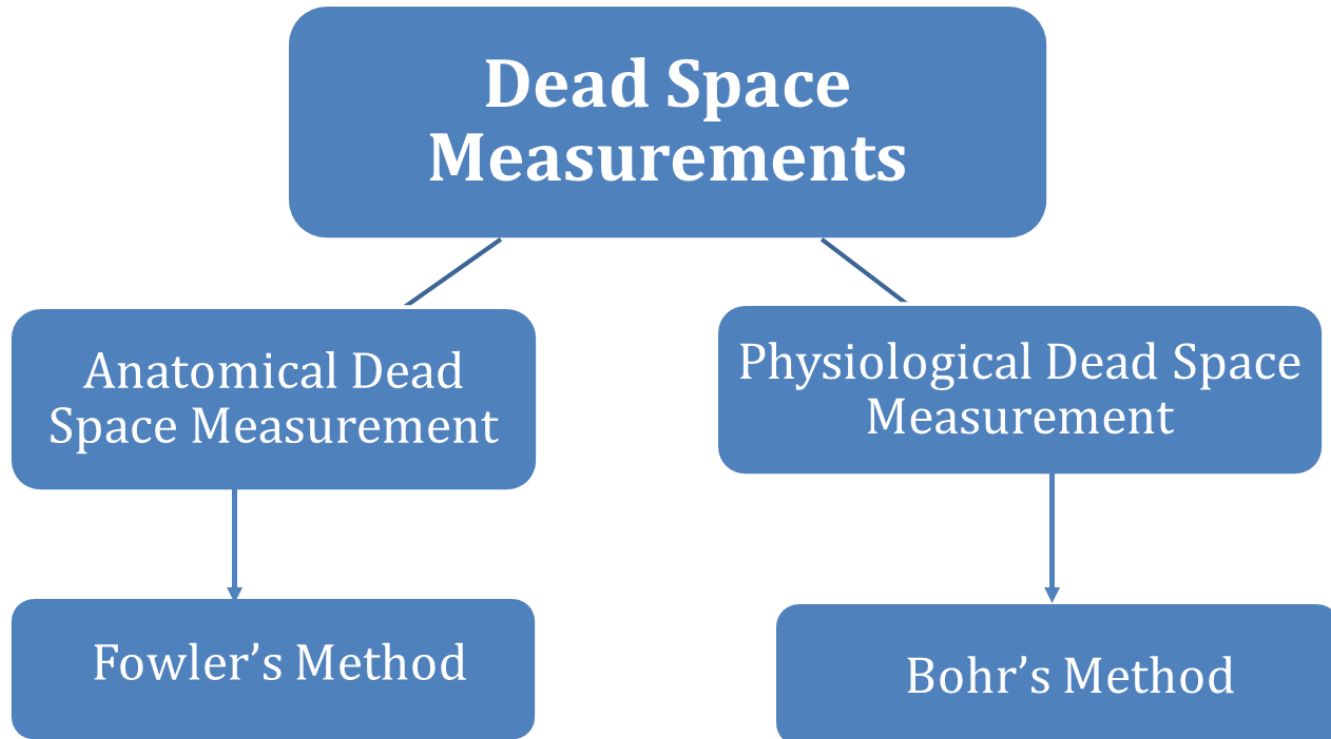
How can you use this formula to calculate V_D ?

Measuring Dead Space

In healthy human adults, dead space is estimated at :

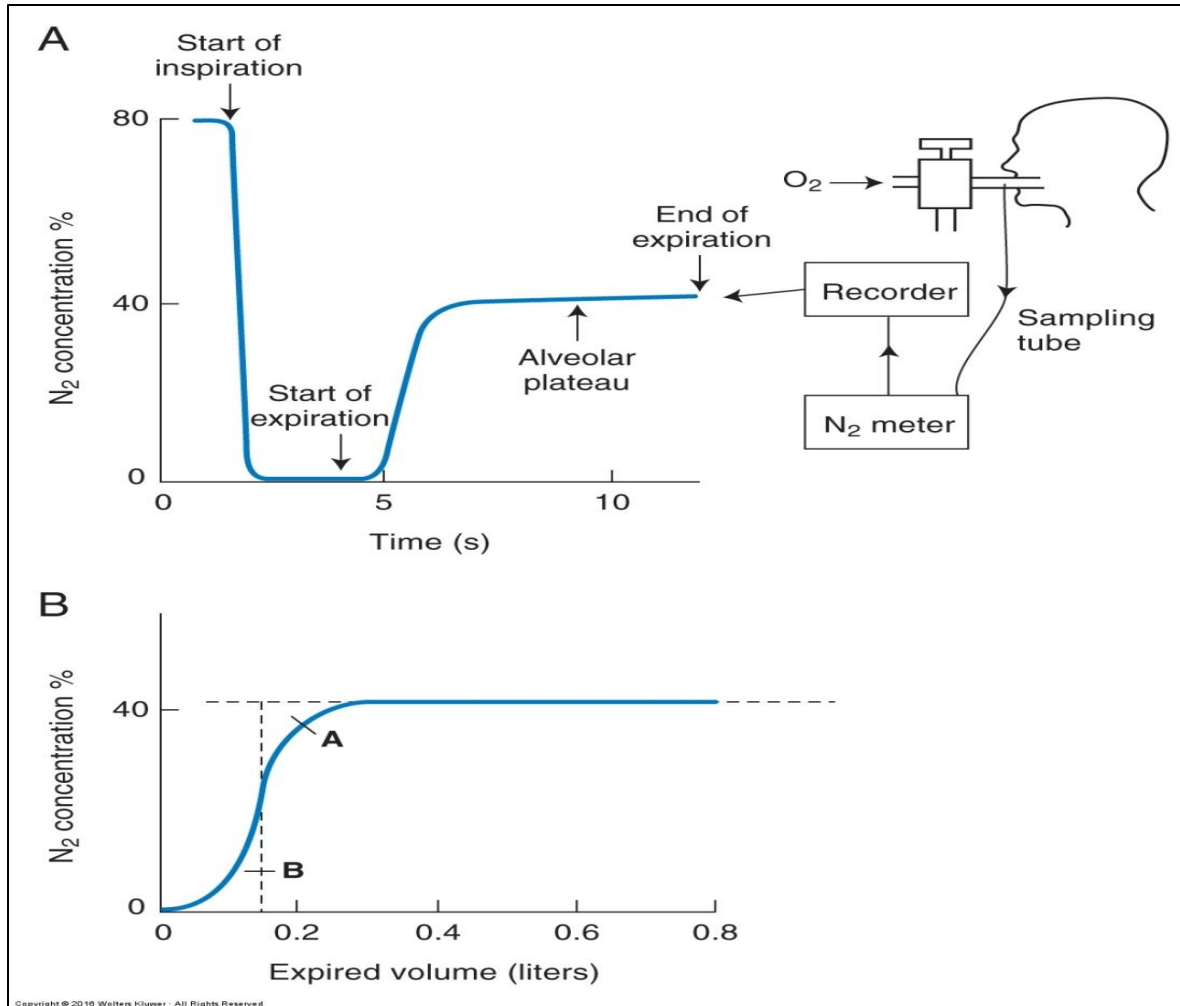
~1 ml for each pound of the subject's weight

Hence wasted ventilation for a healthy 150lb adult is about 150 ml
 $\times 12 \text{ min}^{-1} = 1800 \text{ ml/min}$



Measuring Anatomic Dead Space

Fowler Method



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Anatomical Dead Space Measurement

Fowler Method

- Inspire 100% O_2 after a tidal expiration
- Exhale to RV
- O_2 in dead space not used in gas exchange, therefore expired unaltered as pure O_2 with no N_2
- Gas leaving alveolar region is a mix of O_2 and stale air with N_2
- Expired N_2 volume and concentration are measured

Bohr Method

$$\frac{V_D}{V_T} = 1 - \frac{P_E C O_2}{P_a C O_2}$$

Typical Example

- $P_a C O_2$ – arterial P_{CO_2} (~ 40 mm Hg)
- $P_E C O_2$ – expired P_{CO_2} (~ 28 mm Hg)
- $1 - 28/40 = 1 - 0.7 = 0.3$
- If $V_T = 500$ ml
- $V_D = 0.3 * 500\text{ml} = 150$ ml

Bohr Method

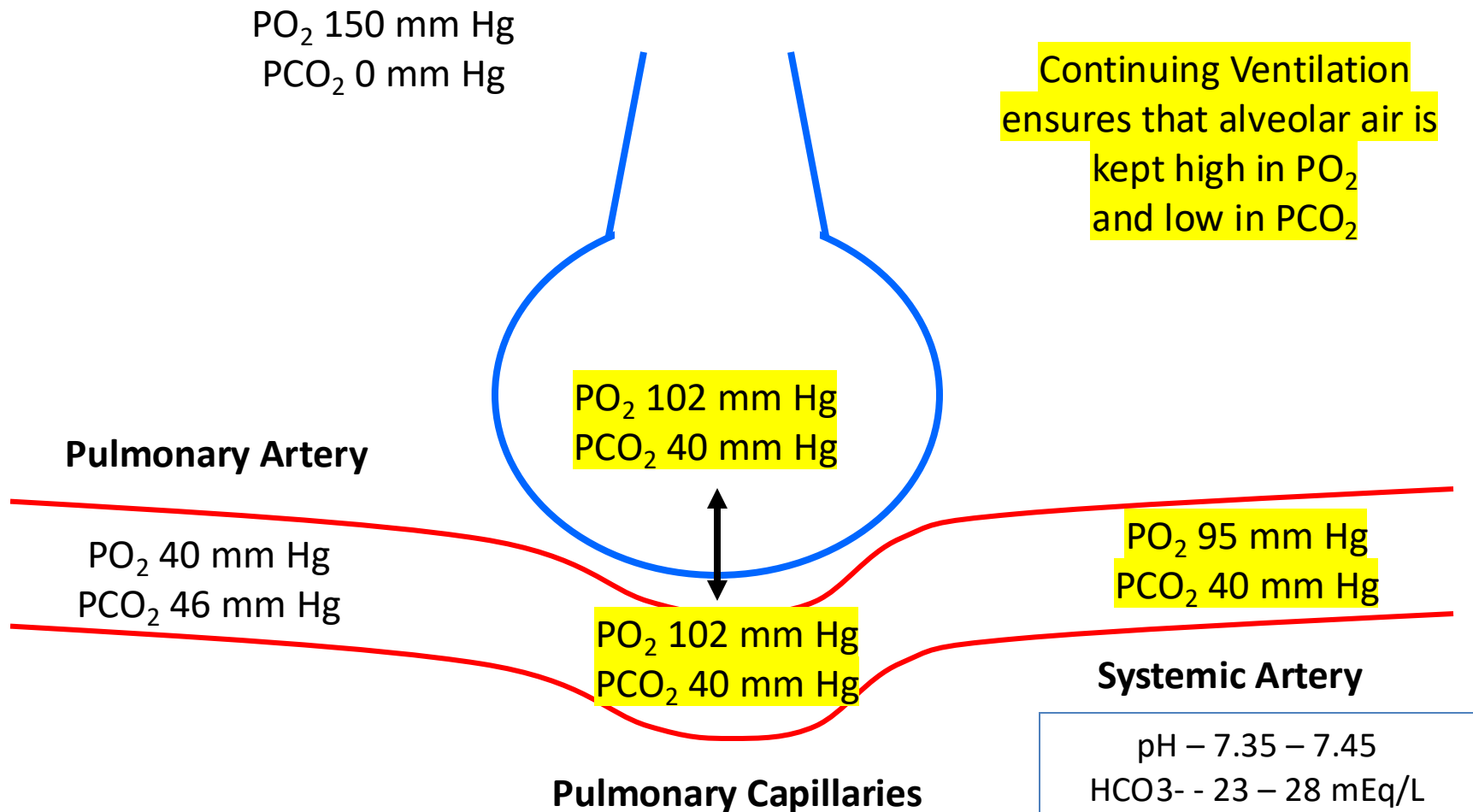
- There is virtually no CO₂ in inspired air
- The CO₂ in expired air must come from alveoli that are both ventilated and perfused
- Inspired air entering anatomical dead space or unperfused alveoli will exit unchanged
- Clinically it is assumed that the alveolar CO₂ and arterial CO₂ are the same, so P_aCO₂ can be used in place of P_ACO₂

$$\frac{V_D}{V_T} = 1 - \frac{P_E CO_2}{P_a CO_2}$$

Fowler vs. Bohr

- Fowler's methods measures **anatomic dead space** only
- **Physiological dead space** and can be measured by the Bohr method.
- E.g. with pulmonary embolism
 - N_2 and O_2 can come from unperfused alveoli (stale air)
 - such stale air will be mixed with expired air throughout alveolar plateau
 - hence Fowler method only measures anatomic dead space, ineffectual for detecting physiological dead space

Constant Ventilation allows maintenance of steady state [gas]



Alveolar Ventilation-

The Effects of **Depth** and **Rate**

Subject	Tidal Vol. (ml)/ Depth	Frequency (breaths min ⁻¹)/ Rate	Minute Ventilation (V _T * f)	Dead Space Ventilation (ml min ⁻¹)	Alveolar Ventilation (ml min ⁻¹)
A	150	40	6000	150*40 (6000)	0
B	500	12	6000	150*12 (1800)	4200
C	1000	6	6000	150*6 (900)	5100

- Alveolar ventilation is dependent on:
 - Depth** of breathing (tidal volume)
 - Rate** of breathing (frequency)
- Increasing the depth of breathing is far more effective in elevating alveolar ventilation than is increasing the frequency or rate of breathing.

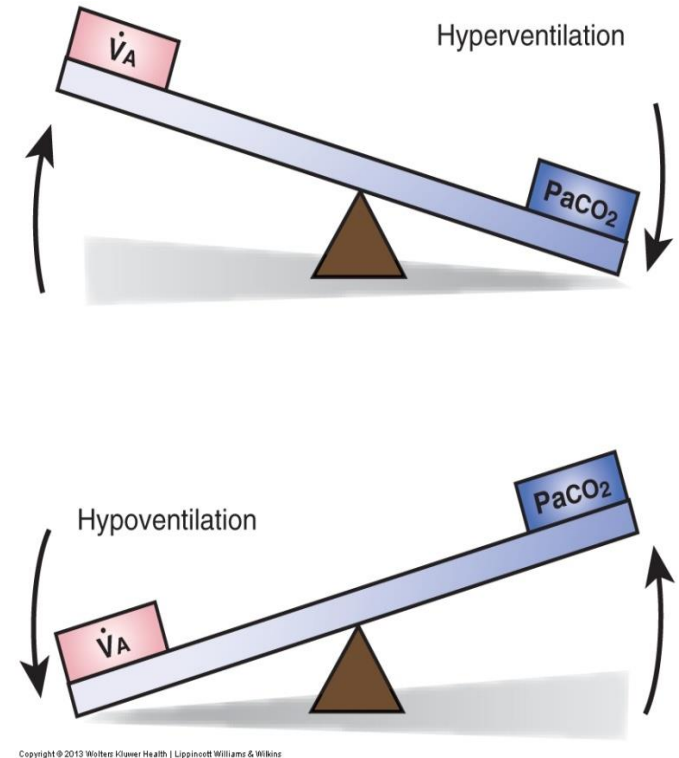
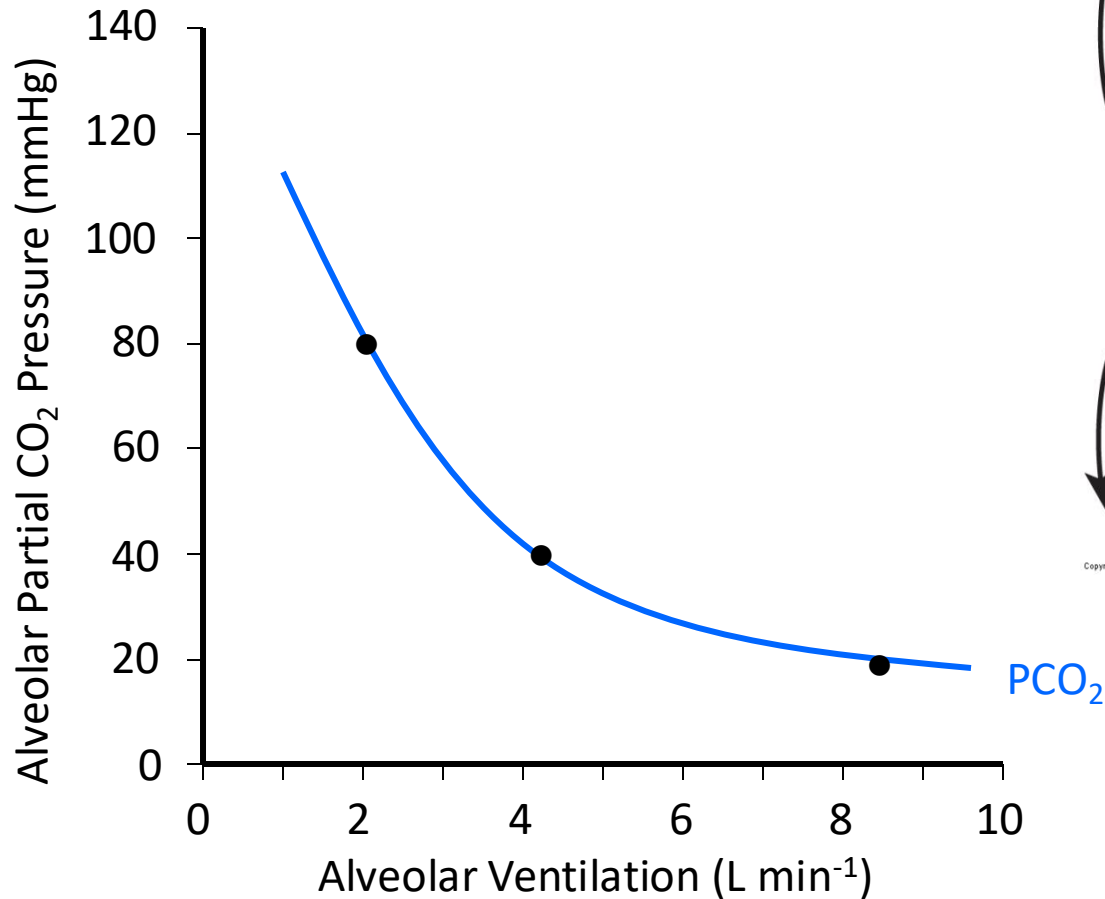
The Alveolar Ventilation Equation based on $P_{A/a}CO_2$

$$\dot{V}_A = \frac{\dot{V}_E CO_2 \times 0.863}{P_A CO_2}$$

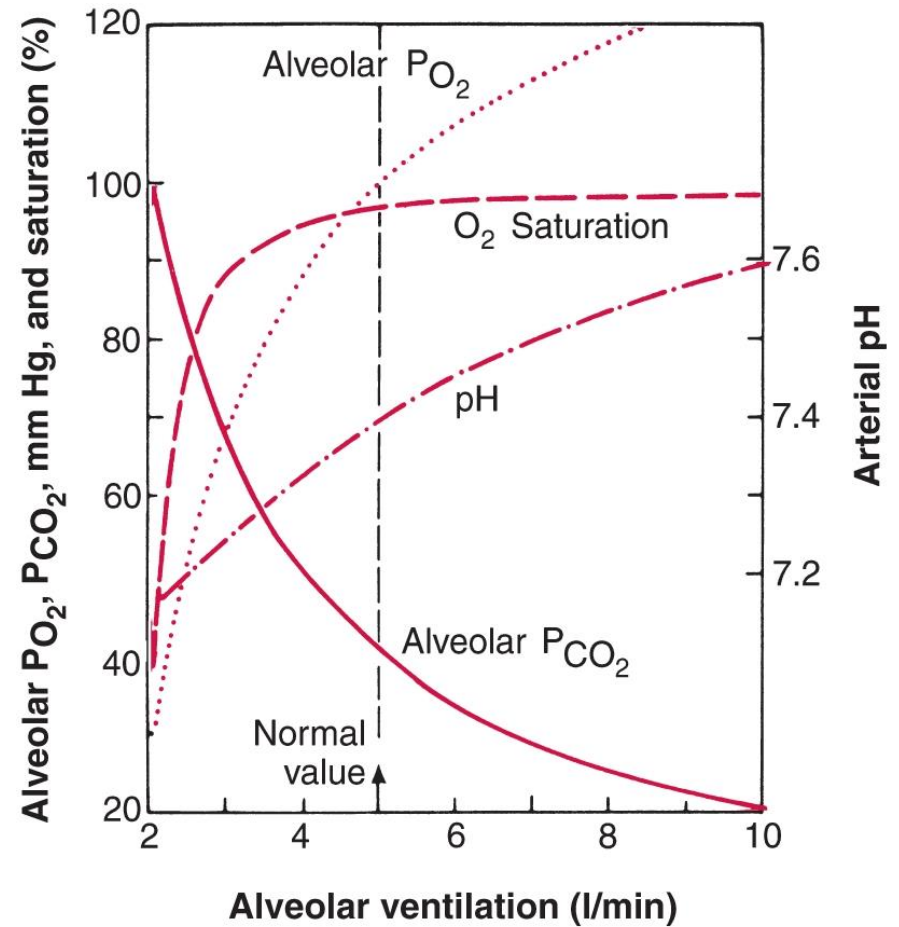
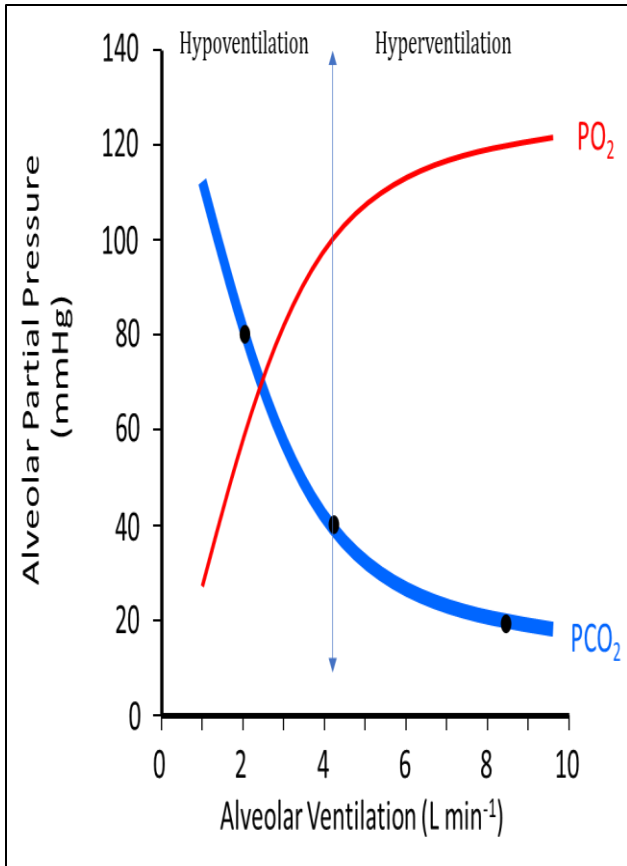
- $\dot{V}_A$ - Alveolar Ventilation
- $\dot{V}_E CO_2$ - Expired volume of CO_2 in 1 minute (200 ml)
- Constant has units mm Hg x L / ml, hence $\dot{V}_A$ has units L / min
- $P_A CO_2$ - partial pressure of alveolar CO_2
- Note: clinically $P_a CO_2$ is used to determine $P_A CO_2$ since $P_a CO_2$ is almost never diffusion limited

Alveolar Ventilation and $P_{A/a}CO_2$

V_A and $P_{a/A}CO_2$ are inversely related.



Alveolar Partial Gas Pressures with Hyper- or Hypo- ventilation



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Alveolar Partial Gas Pressures with Hyper- or Hypo- ventilation

- Alveolar ventilation is usually evaluated in terms of arterial P_{CO_2}
- Increased alveolar ventilation beyond that required to meet metabolic needs is termed **hyperventilation**.
- Hyperventilation leads to **hypocapnia** (P_{aCO_2} is below normal).
- Decreased alveolar ventilation below that required to meet metabolic needs is termed **hypoventilation**.
- Hypoventilation leads to decrease in P_{aO_2} and **hypercapnia** (elevated in P_{aCO_2})

The Alveolar Gas Equation

- Calculates global alveolar partial pressure of oxygen

$$P_A O_2 = P_I O_2 - (P_A CO_2 / R)$$

$P_I O_2 = FIO_2 \times (P_B - P_{H_2O}) = 150$ mm Hg in an healthy individual breathing at sea level

$$P_A O_2 = 150 - (P_a CO_2 / R)$$

$P_a CO_2$ can be substituted for $P_A CO_2$

R – respiratory quotient

≈ 0.8 in normal conditions

Factors determining P_AO_2

Factors that can affect the P_AO_2 are:

- **P_B** - alveolar O_2 is increased with increased atmospheric pressure (hyperbaric chamber) and decreased when there is lowered atmospheric pressure (high altitude).
- **FIO_2** – is 21% in ambient air, this can be altered in the clinical setting when supplemental O_2 is given
- **Metabolic activity** - high metabolic activity uses oxygen and would push P_AO_2 down
- **Ventilation** - increasing ventilation elevates P_AO_2

The A-a O₂ Gradient

- The difference between the alveolar oxygen gas tension ($P_A O_2$) and the arterial oxygen tension ($P_a O_2$).

$$A - a \text{ O}_2 \text{ gradient} = P_A O_2 - P_a O_2$$

- Where $P_A O_2$ is calculated from the alveolar gas equation and $P_a O_2$ is measured in arterial blood.

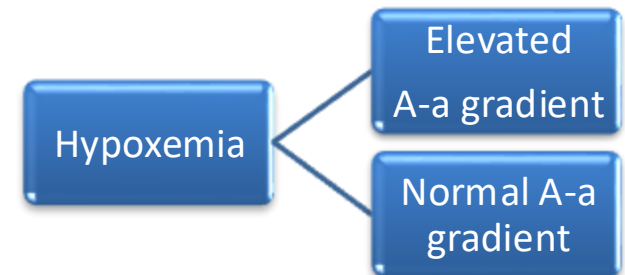
Formula:

$$A - a \text{ O}_2 \text{ gradient} = [P_I O_2 - (P_A C O_2 / R)] - P_a O_2$$

- Normal values of the A-a O₂ gradient are between 5-15 mm Hg

The A-a O₂ Gradient

- A value of more than 15 is usually associated with hypoxemia
- Elevated A-a implies high respiratory effort to achieve oxygenation, i.e. a pulmonary problem
- Low oxygenation due to hypoventilation (i.e. extra-pulmonary source of hypoxemia) would not affect A-a gradient



Definitions of Ventilation terms

- Hypoventilation
 - decreased alveolar ventilation with increased $P_a\text{CO}_2$
 - from Greek 'hypo' meaning 'under'
- Hyperventilation
 - increased alveolar ventilation with decreased $P_a\text{CO}_2$
 - from Greek 'hyper' meaning 'over'
- Hypercapnia
 - increased carbon dioxide in blood ($P_a\text{CO}_2$ above 48 mm Hg)
 - from Greek 'kapnos' meaning 'smoke'
- Hypopnea
 - decreased ventilation in response to lowered metabolic CO_2 production
- Hyperpnea
 - increased ventilation in response to increased metabolic CO_2 production (e.g. exercise), which should purge CO_2

Definitions of Breathing terms

- Eupnea
 - normal breathing
 - from Greek eu - 'well' and pnoia - 'breath'
- Tachypnea
 - increased frequency of breathing ('swift' breath). Ventilation may or may not change depending on tidal volume
- Dyspnea
 - shortness of breath / labored breathing ('ill' breath)
- Apnea
 - temporary cessation of breathing ('a' prefix = without)

Any Questions ?

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